

TeleHand: Hand-only Teleportation for Distant Object Pointing in Virtual Reality

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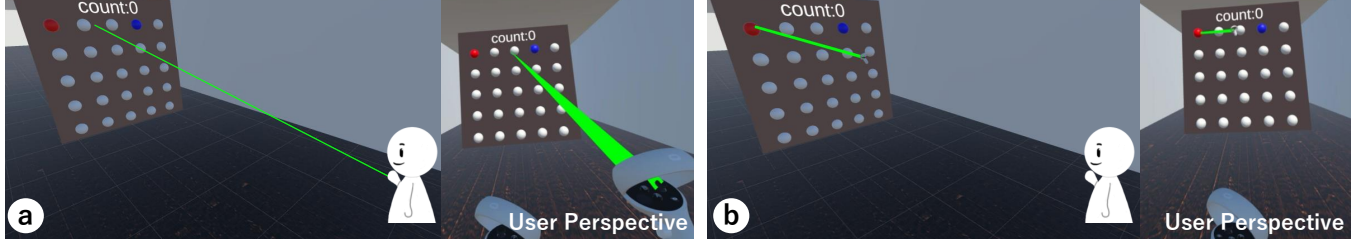


Figure 1: Overview of the TeleHand pointing method. When pointing at a distant red sphere, the user first roughly aims the ray toward the target and presses the “Teleport” button (a). This action teleports only the user’s hand near the target, allowing the user to point at objects using the ray cast from the teleported hand (b).

ABSTRACT

The goal of this study is to present an easy-to-use and accurate pointing technique for distant objects in a virtual reality environment. For this purpose, we propose TeleHand pointing, in which the user first teleports only their hand near a target and points at the target using the ray cast from the teleported hand. To evaluate the usability of our method, we performed a user study in which participants repeated selection tasks for distant targets. As a result, we found that, although the proposed method slightly increases the task completion time, it significantly reduces pointing errors for distant targets compared to the traditional ray-based method.

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1 INTRODUCTION

Object pointing in a virtual reality (VR) environment is a fundamental task. A common pointing method involves casting a linear ray from a handheld controller to intersect with target objects [6]. However, using the ray-based method, pointing at distant objects is challenging because even small hand rotations and tremors can cause significant deviations in the rays around the distant targets.

Various methods have been developed for pointing at distant objects in a VR environment. For instance, Kiyokawa et al. [3] and Han et al. [2] introduced portal-based interactions, allowing users

to place a portal connecting two different points in a VR space and interact with distant objects through the portal. Li et al. [5] proposed vMirror, which enables users to place mirrors in a VR space to interact with objects occluded by other objects using the mirrors. However, these methods require users to carefully place portals or mirrors at positions where target objects can be clearly seen before interaction.

This study aims to present an easy-to-use and accurate pointing technique for distant objects in a VR space. For this, we propose TeleHand pointing, where the user can teleport their hand near a target and point at objects using rays cast from the teleported hand. Pointing at distant objects with a ray is difficult because even small hand rotations and tremors can cause significant deviations in the rays around targets. The proposed method teleports only the hand near the targets, reducing the effects of the hand motions while maintaining the current field of view. This results in easy and accurate pointing for distant targets.

2 USER INTERFACE

In TeleHand pointing, all interactions are performed using a VR controller held in the hand, referred to simply as the “hand.” TeleHand pointing consists of two steps. (Step 1) The user roughly aims the ray cast from their hand toward a target object and presses the “Teleport” button on the controller (Fig 1a). (Step 2) Only the user’s hand is teleported near the target, while its motion remains synchronized with the handheld controller. The user then points at the target using the ray cast from the teleported hand (Fig 1b). The user can also return the teleported hand by pressing the “Return” button. Because our method teleports the hand near the target, it reduces the effect of unintentional hand motions. Additionally, because the method does not teleport the user, they can continue tasks while maintaining their current field of view.

Fig. 2 illustrates the method used to determine the teleportation destination of the hand. When the user presses the “Teleport” button, we generate a probing sphere of radius r , move it along the ray

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extending from the hand, and stop it when it intersects with one of the selectable objects in the scene. Subsequently, we teleport the hand to the center of the probing sphere. We empirically set $r = 20$ m for all experiments in this study.

3 USER STUDY

To evaluate the effectiveness of TeleHand pointing, we conducted a user study comparing it with a traditional method. Fig. 1 shows the VR scene used for the study, where a vertical wall is placed at a distance of D m from the user, and 25 spheres with a radius of 1 m are arranged on the wall.

In this scene, the user performs selection tasks. In each task, two spheres are randomly selected and highlighted in red and blue. The user begins by pointing at the red sphere and pressing the “Selection” button on the controller. They repeat this operation for the blue sphere. The selection task is performed using three different methods: (1) *Ray* (traditional method): The user points at a target using a ray cast from the handheld controller. (2) *Once-TeleHand*: The user teleports their hand near the targets only once during the first selection task. Subsequent selection tasks are performed using the teleported hand. (3) *Each-TeleHand*: The user teleports the hand in each selection task, and the teleported hand is automatically returned to its original position after each selection.

In the user study, participants completed a tutorial session to learn and practice three methods: Ray, Once-TeleHand, and Each-TeleHand. During this session, they repeated the selection task 30 times for each method under the condition that the wall distance $D = 50$ m. Subsequently, in the measurement session, participants repeated the selection task 30 times for each method across four different wall distances ($D = 20, 50, 100$, and 150 m). Note that in the condition with $D = 20$ m, participants performed the task only using the Ray method because the selection targets were sufficiently close.

4 RESULTS AND DISCUSSION

Twelve undergraduate engineering students participated in the user study. Nine had little experience using head-mounted displays (HMDs), while the remaining three regularly used HMDs.

Fig. 3 summarizes the average time taken to complete selection tasks 30 times and the average number of selection errors during the tasks. Each-TeleHand resulted in significantly longer task durations compared to the other methods. This is primarily because of the additional operation required to teleport the hand in the Each-TeleHand method. Alternatively, the proposed methods (Once-TeleHand and Each-TeleHand) significantly reduced the number of selection errors compared to the traditional method. Notably, even for extremely distant targets (i.e., $D = 100$ and 150 m), the proposed methods enabled participants to complete the task with fewer errors. This reduction in errors is attributed to the capability to teleport the hand near distant targets, thereby reducing the impact of unintended minor rotations and tremors of the hand.

We also conducted regression analysis on the selection time and the index of difficulty based on Fitts’s law [1]. We utilized the index of difficulty ID_{DP} presented by Kopper et al. [4] to model pointing operations with a linear ray cast from the handheld controller. Table 1 summarizes the average of the coefficients of determination R^2

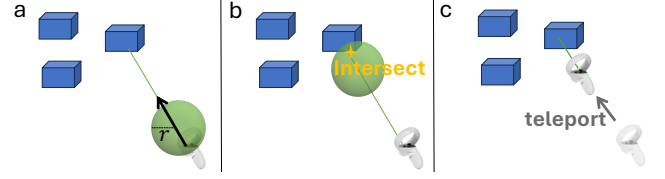


Figure 2: Sphere casting to determine the teleport destination.

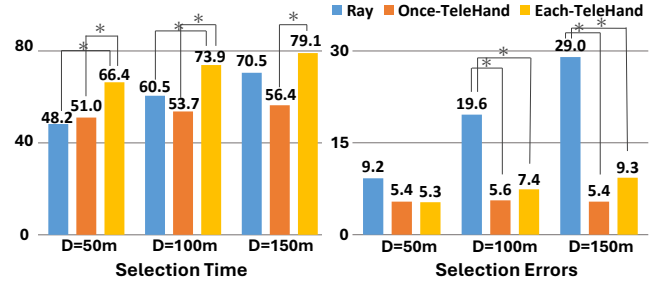


Figure 3: Selection time (left) and error (right) in the tasks. Significant differences between groups are marked with asterisks : * $P < 0.01$.

for the three methods across the four conditions. We observed high coefficients of determination when selecting distant targets using TeleHand ($D = 50$ and 100 , * -TeleHand), comparable to those obtained when selecting nearby targets using the traditional method ($D = 20$, Ray). This suggests that users could select distant targets using the proposed method with a similar sense of ease as when selecting nearby targets using the traditional method.

Table 1: Average coefficients of determination obtained by linear regression analysis on selection time and ID_{DP} .

R^2	D = 20	D = 50	D = 100	D = 150
Ray	0.58	0.38	0.20	0.23
Once-TeleHand	-	0.52	0.43	0.31
Each-TeleHand	-	0.50	0.52	0.40

5 CONCLUSION

This study proposed a method for pointing at distant objects in VR. The key idea is to teleport only the hand near the objects and point using a ray cast from the teleported hand. Our user study showed a significant reduction in pointing errors during distant object selection tasks. Users could select distant objects using our method with a comparable sense of ease as when selecting nearby targets using the traditional method. Future work will focus on automating the determination of the radius of the probing sphere and exploring alternative methods for determining the teleport destination.

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